

Numeration, Division by Zero, and Direct Base Conversion

Throughout, a base is an integer at least 2. A base tells us how an integer is represented; it does not change the integer itself.

1. Exercise 1.15: parity in different bases

(a) In which bases is $(11)_b$ even? By place value, $(11)_b = b + 1$. Thus any odd base works. For example,

$$(11)_3 = 4, \quad (11)_5 = 6, \quad (11)_7 = 8.$$

(b) Conjecture and proof. The number $(11)_b$ is even if and only if b is odd. Indeed, $(11)_b = b + 1$, and adding 1 reverses parity.

(c) A general parity test. Let

$$N = (a_n a_{n-1} \cdots a_1 a_0)_b = a_n b^n + \cdots + a_1 b + a_0,$$

where $0 \leq a_i < b$.

- If b is **even**, every term except a_0 is even. Consequently, N is even exactly when its last digit is even.
- If b is **odd**, every power of b is odd, so $a_i b^i$ has the same parity as a_i . Consequently, N is even exactly when its digit sum $a_n + \cdots + a_0$ is even.

Here “digit sum” means the sum of the numerical values of the digits.

2. Why $(121)_b$ is a perfect square

For every integer base $b \geq 3$,

$$(121)_b = b^2 + 2b + 1 = (b + 1)^2.$$

Since $b + 1$ is an integer, this is a perfect square. The condition $b \geq 3$ ensures that the digit 2 is permitted.

This proof needs no reference to Exercise 1.15. As an optional observation, $b + 1 = (11)_b$, so $(121)_b = ((11)_b)^2$.

$$(121)_3 = 16 = 4^2, \quad (121)_5 = 36 = 6^2, \quad (121)_{10} = 121 = 11^2.$$

3. Division by zero

The proposed rules in the exercise are $n/0 = n$, associated with Mahavira, and $n/0 = \infty$, associated with Bhaskara II. The motivations below are possible intuitive explanations, not claims about their documented reasoning.

(a) Why might someone propose $n/0 = n$? One possible intuition is that dividing into “zero groups” means doing nothing to the original quantity. Another is an incorrect analogy with division by 1, which leaves a number unchanged. Neither intuition satisfies the definition of division.

(b) Why does $n/0 = n$ contradict ordinary arithmetic? Division reverses multiplication: a quotient $n/d = q$ must satisfy $dq = n$. If $n/0 = n$, then

$$0 \cdot n = n.$$

But $0 \cdot n = 0$, so this would force $n = 0$. Taking $n = 5$ gives the contradiction $0 = 5$. Thus the proposed rule fails for every nonzero n .

This argument uses the definition of a quotient, rather than assuming we may cancel an undefined expression such as $(n/0) \cdot 0$.

(c) Why might someone propose $n/0 = \infty$? For a positive numerator, division by progressively smaller positive numbers gives progressively larger quotients:

$$\frac{1}{1} = 1, \quad \frac{1}{0.1} = 10, \quad \frac{1}{0.01} = 100, \quad \frac{1}{0.001} = 1000.$$

This suggests unbounded growth as the divisor approaches zero *from above*. It does not give a value at zero. Approaching zero from below instead produces negative quotients of unbounded magnitude.

(d) Why does $n/0 = \infty$ cause arithmetic problems? Suppose a single new object ∞ is the quotient for every positive numerator, and suppose division still reverses multiplication. Then

$$1/0 = \infty \implies 0 \cdot \infty = 1, \quad 2/0 = \infty \implies 0 \cdot \infty = 2.$$

The same product would equal both 1 and 2, a contradiction. Such an extension therefore cannot preserve the ordinary rules of arithmetic.

What about $0/0$? The equation $0q = 0$ holds for every q , so it selects no unique quotient. For a nonzero numerator, $0q = n$ has no solution; for a zero numerator, it has no unique solution. Division by zero is undefined in ordinary real arithmetic.

4. Direct conversion between bases

A general method: repeated division

To convert a base-5 numeral to base 6, repeatedly divide its value by 6, carrying out the arithmetic in base 5. Since $6 = (11)_5$, this is base-5 division by $(11)_5$.

At each step write $N = 6Q + r$, with $0 \leq r < 6$. The remainder is the next base-6 digit, starting at the units place. Repeat with Q until the quotient is zero, then read the remainders in reverse order. A remainder of value five is written $(10)_5$ during base-5 arithmetic, but becomes the single digit 5 in the base-6 answer.

This works between any two integer bases. No intermediate decimal numeral is required.

An algebraic method: substitution and regrouping

For small examples, we can expand the original place-value expression in powers of the new base. Then carry or borrow until every coefficient is a legal digit. **Example 1:** $(123)_5$ to base 6. Since $5 = 6 - 1$,

$$\begin{aligned}(123)_5 &= 5^2 + 2 \cdot 5 + 3 \\ &= (6 - 1)^2 + 2(6 - 1) + 3 \\ &= 6^2 - 2 \cdot 6 + 1 + 2 \cdot 6 - 2 + 3 \\ &= 6^2 + 2.\end{aligned}$$

Thus $\boxed{(123)_5 = (102)_6}$.

Example 2: $(123)_5$ to base 3. Since $5 = 3 + 2$,

$$\begin{aligned}(123)_5 &= (3 + 2)^2 + 2(3 + 2) + 3 \\ &= 3^2 + 6 \cdot 3 + 11.\end{aligned}$$

The coefficients are too large to be base-3 digits. Carry using $11 = 3 \cdot 3 + 2$:

$$\begin{aligned}3^2 + 6 \cdot 3 + 11 &= 3^2 + 9 \cdot 3 + 2 \\ &= 3^3 + 3^2 + 2.\end{aligned}$$

Thus $\boxed{(123)_5 = (1102)_3}$.

Example 3: $(123)_5$ to base 7. Since $5 = 7 - 2$,

$$\begin{aligned}(123)_5 &= (7 - 2)^2 + 2(7 - 2) + 3 \\ &= 7^2 - 4 \cdot 7 + 4 + 2 \cdot 7 - 4 + 3 \\ &= 7^2 - 2 \cdot 7 + 3.\end{aligned}$$

The negative coefficient is not a legal digit. Regroup one 7^2 as seven copies of 7:

$$7^2 - 2 \cdot 7 + 3 = (7 - 2) \cdot 7 + 3 = 5 \cdot 7 + 3.$$

Thus $\boxed{(123)_5 = (53)_7}$.

All three conversions describe the same integer:

$$\boxed{(123)_5 = (102)_6 = (1102)_3 = (53)_7}.$$

5. In what base are we doing the algebra?

The operations are ordinary integer algebra. The written numerals still use a notation convention.

In an expression such as

$$(123)_5 = 5^2 + 2 \cdot 5 + 3,$$

the unsubscripted numerals are written in our usual decimal notation. The symbol 5 denotes the integer five; the equality concerns integers, regardless of how we represent them.

Substituting $5 = 7 - 2$ and expanding uses base-independent algebraic identities:

$$(7 - 2)^2 + 2(7 - 2) + 3 = 7^2 - 2 \cdot 7 + 3.$$

This expression is not yet a base-7 numeral: its coefficients include a negative number. Only after regrouping it as $5 \cdot 7 + 3$ do the coefficients become the digits of $(53)_7$.

There are therefore two stages:

1. **Algebra:** express the value in powers of the target base, using ordinary integer identities.
2. **Representation:** normalize the coefficients by carrying or borrowing so that each lies between 0 and $b - 1$.

These examples avoid computing the entire value as an intermediate decimal numeral. They do still write intermediate coefficients, such as 11 in the base-3 example, in decimal notation. If every written intermediate numeral must be in the source or target base, the notation must be adjusted accordingly.

*A base is a way of representing an integer,
not a different arithmetic in which the integer lives.*